# Time-Expanded Graph Learning for Multi-Agent Temporal Interaction Modeling with Counterfactual Interventions

Vincenzo Marco De Luca[1], Antonio Longa[2], Giovanna Varni[1], Andrea Passerini[1]

[1]Department of Information Engineering and Computer Science, University of Trento, Italy
[2]UiT The Arctic University of Norway

Graph Neural Networks for real-world multi-agent systems require benchmarks that jointly capture relational structure, temporal evolution, and interaction quality. However, most existing graph datasets remain either static or low-stakes, failing to reflect coordination breakdowns in realistic, safety-critical environments.

In this work, we address this gap by introducing a modeling framework grounded in a real-world high-stakes operating room setting, where team interactions directly influence procedural safety and efficiency. Unlike standard benchmarks, our setting provides structured supervision over teamwork quality and failure modes, enabling socially-aware graph learning. Crucially, we leverage this rich, multi-granular supervision to mitigate the data scarcity and high collection costs inherent to clinical environments, enabling data-efficient learning.

We build upon a multimodal operating room dataset [3] and define a multi-level annotation scheme designed to capture complementary aspects of team dynamics. Each six-minute video segment is annotated with: **(i)** 16 node-level behavioral descriptors capturing individual teamwork traits (e.g., leadership, situational awareness, decision-making), **(ii)** 5 edge-level interaction descriptors modeling communication, coordination, and interaction patterns, and **(iii)** 6 graph-level labels capturing overall team performance and procedural duration.

This results in a multi-level supervision framework, beyond standard action- or phase-based annotations, that maps directly to core graph learning tasks: node classification for individual traits, link classification for interaction patterns, and graph classification for overall team efficiency.

A key contribution is the counterfactual annotation layer, which provides a human-grounded benchmark for causal reasoning on graphs. Moving beyond descriptive scoring, expert annotators identify the most critical interaction-level event responsible for degradation in teamwork quality. Each annotation specifies: (i) temporal interval, (ii) responsible agent(s), (iii) interaction type (e.g., missed acknowledgment, delayed response, unclear directive, coordination breakdown), and (iv) affected teamwork dimension (communication, coordination, leadership, situational awareness, cooperation/back-up behavior). This enables localization of not only *that* performance degraded, but also *where and why*.

To support learning and interpretability, each counterfactual is enriched with structured descriptors: criticality, plausibility, confidence, minimality, and expected improvement. These attributes enable prioritization of failure causes according to impact and reliability, bridging observational annotation and intervention-oriented reasoning.

To model these dynamics, we propose a time-expanded graph representation [1], transforming temporal multi-agent interactions into a unified, heterogeneous time-expanded graph, where typed edges encode both structural and temporal dependencies. While computationally scalable due to the bounded interaction windows, this formulation allows standard GNNs to jointly reason over who interacts with whom and when, bypassing the recurrent constraints of Temporal GNNs, thereby improving interpretability and reducing fragmentation sensitivity.

Building on this representation, we study predictive and intervention-based objectives, including surgical efficiency prediction (efficient vs. inefficient procedures) and multi-level counterfactual reasoning aimed at identifying minimal structural perturbations at the node and edge levels that maximize expected performance improvements. We explore greedy [1] and factual explanation-based strategies [2] for minimal performance-improving interventions.

Overall, this work bridges graph representation learning, social behavior modeling, and counterfactual reasoning in high-stakes environments, providing a generalizable framework for predicting, explaining, and intervening on team dynamics. Ultimately, this work bridges foundational graph representation learning with actionable, accountable AI in safety-critical multi-agent systems.

## References

[1] V. M. De Luca, A. Longa, G. Varni, and A. Passerini. Actionable real-time modeling of surgical team dynamics via time-expanded interaction graphs. *arXiv preprint arXiv:2605.04169*, 2026. Accepted at HHAI 2026.

[2] V. M. De Luca, A. Longa, G. Varni, and A. Passerini. From multimodal observation to interpretable suggestions: Counterfactual time-expanded relational modeling of surgical teams. 2026. Under submission.

[3] E. Özsoy, C. Pellegrini, T. Czempiel, F. Tristram, K. Yuan, D. Bani-Harouni, U. Eck, B. Busam, M. Keicher, and N. Navab. Mm-or: A large multimodal operating room dataset for semantic understanding of high-intensity surgical environments. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 19378–19389, 2025.